provides the frequency response of the RC low-pass filter as magnitude and phase (i.e., polar form):

$$Mag\,H(\omega) = \frac{\alpha}{[\alpha^2 + \omega^2]^{1/2}}$$

$$Phase\,H(\omega) = \arctan\left[-\frac{\omega}{\alpha}\right]$$

Figure 13-9 shows graphs of these curves for a cutoff frequency of 1000 hertz (i.e., $\alpha = 2\pi 1000$).

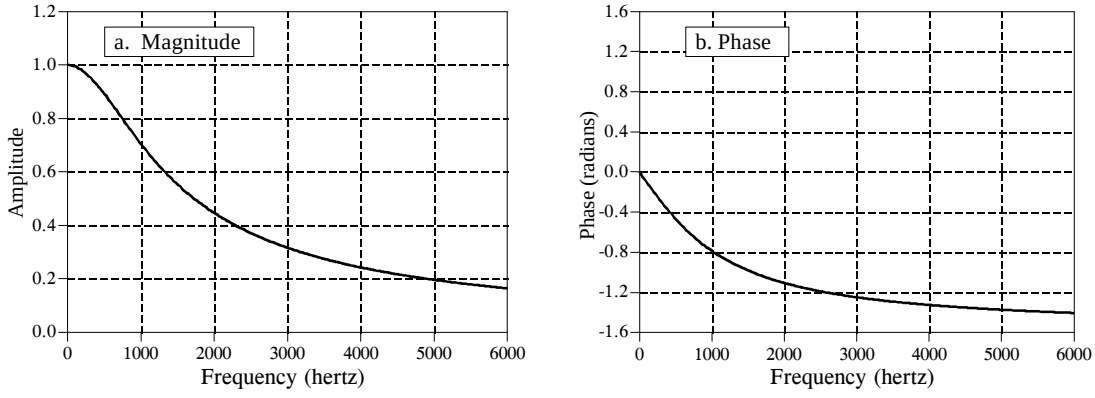

FIGURE 13-9
Frequency response of an RC low-pass filter. These curves were derived by calculating the Fourier transform of the impulse response, and then converting to polar form.

The Fourier Series

This brings us to the last member of the Fourier transform family: the *Fourier series*. The time domain signal used in the Fourier series is *periodic* and *continuous*. Figure 13-10 shows several examples of continuous waveforms that repeat themselves from negative to positive infinity. Chapter 11 showed that periodic signals have a frequency spectrum consisting of **harmonics**. For instance, if the time domain repeats at 1000 hertz (a period of 1 millisecond), the frequency spectrum will contain a first harmonic at 1000 hertz, a second harmonic at 2000 hertz, a third harmonic at 3000 hertz, and so forth. The first harmonic, i.e., the frequency that the time domain repeats itself, is also called the **fundamental frequency.** This means that the frequency spectrum can be viewed in two ways: (1) the frequency spectrum is *continuous*, but zero at all frequencies except the harmonics, or (2) the frequency spectrum is *discrete*, and only *defined* at the harmonic frequencies. In other words, the frequencies between the harmonics can be thought of as having a value of zero, or simply